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Cobalt-catalyzed Biaryl Couplings *via* C—F Bond

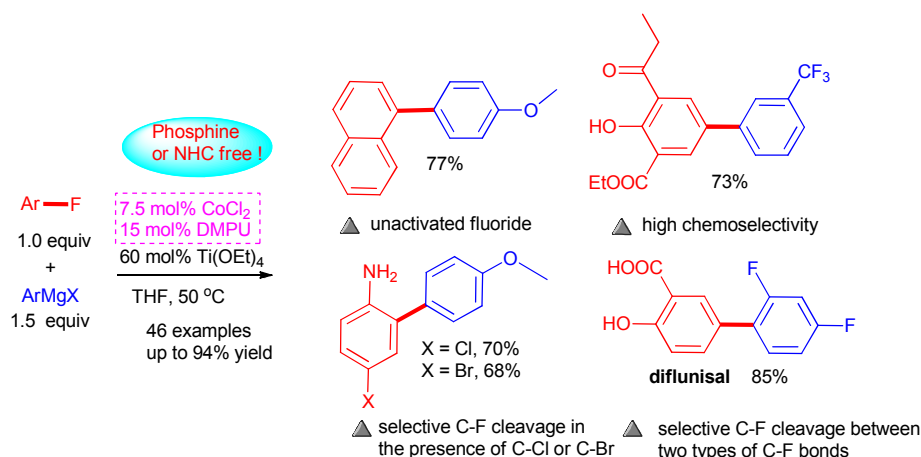
Activation in the Absence of Phosphine or NHC Ligands

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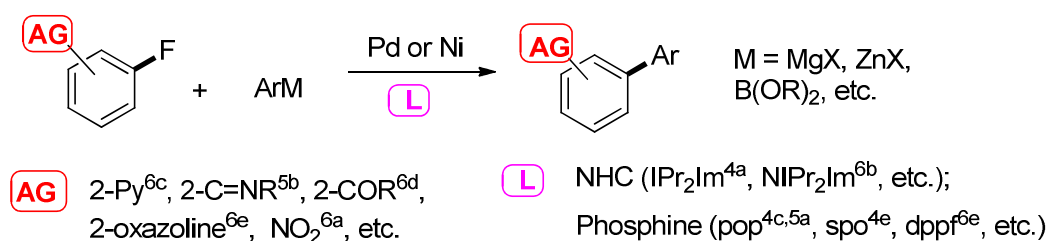


ABSTRACT. A highly general and selective Co-catalyzed biaryl coupling through C—F cleavage under phosphine or NHC free conditions was described. A broad range of aryl fluorides including unactivated fluorides as well as those with sensitive functionalities could couple with various Ti(OEt)₄-mediated aryl Grignard reagents with high selectivity under the catalysis of CoCl₂/DMPU. Importantly, selective C—F bond activation couplings between two types of fluorines (difluorinated aromatics and on two different coupling partners) and in the presence of C—Cl or C—Br bonds could also be achieved.

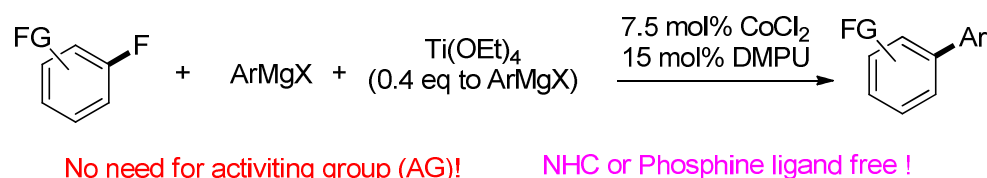
1. INTRODUCTION

Fluorine-containing organic compounds have become increasingly important in a wide range of areas such as medicine, agrochemistry, catalysis, materials science and biochemistry.¹ The development of undocumented reactions for functionalization of carbon-fluorine bonds are of high significance² because: a) the reactions will help in understanding of the activation of highly stable bonds and in searching for highly reactive catalyst systems; b) selective carbon-carbon formation couplings *via* C–F cleavage of polyfluorinated molecules also provide an attractive way to make new fluorine-containing compounds;² and c) the conversions of fluorides *via* C–F cleavage to versatile building blocks such as organoborons are a stimulating way to derivatize the organofluorides.³ In this context, numerous transition metal-catalyzed cross-couplings of fluorinated compounds with Grignard reagents,⁴ organozinc reagents⁵ and organoboron reagents⁶ have been developed recently. Similar to the coupling reactions of aryl iodides, bromides, chlorides, and triflates, the catalysts in known couplings of C–F cleavage are mainly dominated by palladium and nickel complexes.² Great efforts have been devoted to developing or

A) Two means to activate C–F bonds (activation groups or special ligands) in the reported Pd/Ni-catalyzed couplings^{2d}



B) This work: Cobalt-catalyzed biaryl couplings via C–F bond activation



Scheme 1 Two means to activate C-F bonds in the reported Pd/Ni-catalyzed couplings and Co-catalyzed C-F activation biaryl couplings in this work

identifying highly effective phosphine^{4c,4e,5a,6c} or NHC^{4a,6b} ligands for Pd or Ni catalyst to activate the highly stable C-F bonds of aryl fluorides (Scheme 1A). On the other hand, a series of functional groups such as 2(4)-NO₂,^{6a} 2-pyridyl,^{6c} 2-C=NR,^{5b} 2-C(R)=O,^{6d} 2-oxazoline^{6e} *etc.* can usually function as an activating group to help to split the C-F bonds in the related Pd or Ni-catalyzed coupling reactions (Scheme 1A). Up to now the reports on Pd or Ni-catalyzed biaryl couplings between aryl fluorides and aryl metal reagents without using the above-mentioned two means (activating groups and phosphine/NHC ligands) were rare.

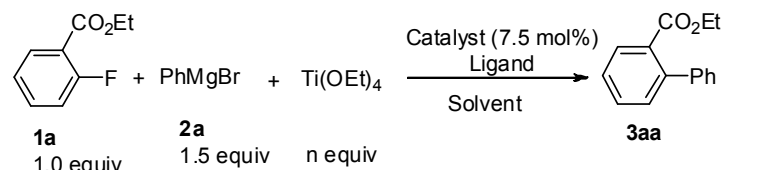
Recently various Co-catalyzed cross-coupling reactions have been intensely and extensively investigated.⁷ Although Co-mediated C-F bond activation in a stoichiometric manner has been reported,⁸ there is only one report concerning the Co-catalyzed cross-couplings of aryl fluorides with aryl metal reagents, where an *ortho* carbonyl group in the fluorides functioned as a key activating group.⁹ To the best of our knowledge, there has been no report on Co-catalyzed biaryl couplings between aryl fluorides and aryl metal reagents in the absence of both activating groups and phosphine/NHC ligands to date. Herein, we report a general and facile Co-catalyzed cross-couplings of aryl fluorides with aryl Grignard reagents in the presence of substoichiometric Ti(OEt)₄ (40 mol % to ArMgX) (Scheme 1B). Salient features of this C-F bond activation reaction lie in the elimination of the need for phosphine/NHC ligands or activating groups. Importantly, selective C-F bond cleavage couplings in the presence of C-Cl or C-Br bonds and between two types of fluorines (difluorinated aromatics and on two different coupling partners) could also be achieved.

2. RESULTS AND DISCUSSION

We started our study with the optimization of reaction conditions using the model cross coupling reaction of ethyl 2-fluorobenzoate (**1a**) with PhMgBr (**2a**). The results were outlined in Table 1. It can be seen from Table 1 that the Pd-catalyzed reactions hardly gave the desired product while the Ni-

catalyzed reaction could occur in a 32% yield (Entries 1-3). The reactions catalyzed by $\text{CoCl}_2/\text{PR}_3$ (R = Me, n-Bu and Cy) proceeded well to afford the product in over 80% yields (Entries 10-12).^{10a} With no

Table 1 Optimization studies^a



Entry	Catalyst/Ligand ^b	n (equiv)	Solvent	T (°C)	Yield (%)
1	$\text{NiCl}_2/\text{PBu}_3$	0.6	THF	50	32
2	$\text{Pd}(\text{PPh}_3)_4$	0.6	THF	50	trace
3	$\text{PdCl}_2/\text{PBu}_3$	0.6	THF	50	trace
4	CoCl_2	0.6	THF	50	12
5	$\text{CoCl}_2/\text{isoquinoline}$	0.6	THF	50	7
6	$\text{CoCl}_2/\text{bipyridine}$	0.6	THF	50	19
7	$\text{CoCl}_2/\text{tmeda}$	0.6	THF	50	25
8	$\text{CoCl}_2/\text{PPh}_3$	0.6	THF	50	32
9	$\text{CoCl}_2/\text{dppp}$	0.6	THF	50	58
10	$\text{CoCl}_2/\text{PBu}_3$	0.6	THF	50	84
11	$\text{CoCl}_2/\text{PMe}_3$	0.6	THF	50	80
12	$\text{CoCl}_2/\text{PCy}_3$	0.6	THF	50	82
13 ^c	CoCl_2	0.6	DMPU/THF	50	83
14 ^c	CoCl_2	0.6	NMP/THF	50	21
15	CoCl_2	0.6	DMPU/ether	50	35
16	CoCl_2	0.6	DMPU/PhMe	50	29
17	CoCl_2	0.6	DMPU/THF	25	48
18	CoCl_2	0.6	DMPU/THF	reflux	78
19 ^d	CoCl_2	0	DMPU/THF	50	11
20 ^d	CoCl_2	0.3	DMPU/THF	50	30
21	CoCl_2	0.9	DMPU/THF	50	82
22	CoCl_2	1.5	DMPU/THF	50	67
23 ^e	CoCl_2	0.6	DMPU/THF	50	85

^aThe reaction was carried out on a 2 mmol scale. ^bUnless indicated otherwise, the catalyst metals were charged in 7.5 mol % with 15 mol % (monodentate), 7.5 mol % (bidentate) ligand or 15 mol % TMEDA, DMPU and NMP.

^cIncreasing the amount of DMPU and NMP from 15 mol% to 100 mol% gave the similar results. ^dIn these reactions, the products of the addition of PhMgBr to the ester group were found. ^eThe mixture of 1.5 equiv of PhMgBr with 0.6

equiv of $\text{Ti}(\text{OEt})_4$ was added in two portions, 1.0 equiv was added at first, after two-hour reaction at 50 °C, the remaining 0.5 equiv was added.

or other selected ligands, the Co-catalyzed reactions gave unsatisfactory results (Entries 4-9). To our delight, 7.5 mol% CoCl_2 with 15 mol% of DMPU (1,3-dimethyl-3,4,5,6-tetrahydro-2(1H)-pyrimidinone)^{10b,c} could catalyze the reaction equally well with those using PR_3 ; whereas NMP as a co-solvent gave a very low yield (Entries 13 and 14). A transition-metal-catalyzed C–F activation reaction in the absence of NHC or phosphine ligands is noteworthy, therefore we focused on further optimization of the reaction conditions using $\text{CoCl}_2/\text{DMPU}$. Other solvents such as ether, toluene gave disappointing yields (Entries 15 and 16). The reaction at room temperature could not proceed to completion and afforded the product only in a 48%, while the reaction at a higher temperature did not give an improved yield (Entries 17 and 18). The amount of $\text{Ti}(\text{OEt})_4$ had a remarkable influence on the reaction: when $\text{Ti}(\text{OEt})_4$ was loaded below 0.6 equivalent (40 mol% based on the amount of PhMgBr), the products resulted from the addition of PhMgBr to the ester group were formed and the yield of cross-coupling product was rather low; on the other hand, increasing the amount of $\text{Ti}(\text{OEt})_4$ over 0.9 equivalent (60 mol% to PhMgBr) also resulted in a lowered yield (Entries 19-22). It is worth mentioning that titanate complexes $[\text{RTi}(\text{OR})_4 \cdot \text{M}]$, $\text{M} = \text{Li}$ or MgX have been used as a simple means to modify the reactivity of Grignard or lithium reagents for a long time, however, their structures have not been ascertained.^{10a,11}

The species of (hetero)aryl Grignard reagents tuned by substoichiometric titanates (40 mol% to ArMgBr) also exhibit reactivity and selectivity distinct from Grignard reagents^{10a} while their structures wait for further elucidation. Besides, it was found that adding the mixture of PhMgBr and $\text{Ti}(\text{OEt})_4$ in two portions would facilitate the stirring and resulted in a yield rise (to 85%, Entry 23). This improvement became even more obvious in the reactions when active-hydrogen-containing substrates (for example **1f**, **1g**, **1t-1x** in Table 2) were used.

With optimized reaction conditions in hand, we then investigated the generality of this cobalt-catalyzed C–F activation coupling. The results were outlined in Table 2. This phosphine or NHC-free Co-catalyzed biaryl coupling *via* C–F bond cleavage proved to be quite general and could go to

completion in 6-8 hours. Without an activating group, unactivated aryl fluorides such as PhF, 4-MeOC₆H₄F, 1-C₁₀H₇F, *m*-C₆H₄F₂ could all couple with various aryl Grignard reagents smoothly (Table

Table 2 Co-catalyzed C–F activation biaryl couplings under phosphine or NHC free conditions^a

$\text{Ar}-\text{F} \quad \text{1} \quad + \quad \text{Ar}'\text{MgBr} \quad \text{2} \quad + \quad \text{Ti}(\text{OEt})_4$ $1.0 \text{ equiv} \quad 1.5 \text{ equiv} \quad 0.6 \text{ equiv}$				$\xrightarrow[\text{THF, 50 } ^\circ\text{C}]{7.5 \text{ mol\% CoCl}_2, 15 \text{ mol\% DMPU}}$ $\text{Ar}-\text{Ar}' \quad \text{3}$			
Entry	ArF	Ar'	Product	Entry	ArF	Ar'	Product
1		4-MeOC ₆ H ₄ 2b	 3bb 65%	15		2a^d	 3ia 75%
2	1b	4-EtOOC ₆ H ₄ 2c^b	 3bc 55%	16		2d^{b,d}	 3je 72%
3	1b	2-NCC ₆ H ₄ 2d^b	 3bd 55%	17		2a	 3ka 90%
4	1b	2-Py 2e^b	 3be 65%	18		2a	 3la 65%
5		Ph 2a	 3bb 50% (78%) ^c	19		4-MeC ₆ H ₄ 2g	 3mg 85%
6		2b	 3db 77%	20		2b	 3nb 82%
7	1d	4-Me ₂ NC ₆ H ₄ 2f	 3df 70%	21		3-F ₃ CC ₆ H ₄ 2i	 3oi 63%
8		3-MeOC ₆ H ₄ 2g	 3eg 65%	22		2-thiophenyl 2j	 3pj 67%
9	1e	2b^d	 4eb 79%	23		2b	 3qb 75%
10		2b^c	 3fb 79%	24		2i	 3ri 76%
11		2b	3fb 72%	25		3-Py 2k	 3sk 76%
12		2h^c	 3gh 91%	26		2d^d	 3td 92%
13		2h	3gh 89%	27		2a^d	 3ua 86%
14		2a^d	 3v	28		1-Naphthyl 2l^d	 3vl 78%

3ha 88%

Table 2 (Continued)

Entry	ArF	Ar'	Product	Entry	ArF	Ar'	Product
29		2b ^d		32		2a	
30		2b ^d		33		2c	
31	1x	2i ^d		34		2a	

^aAll reactions were carried out on a 2 mmol scale. ^bPyridyl and aryl Grignard reagents with FG (CN, COOEt, etc.) were prepared *via* bromine or iodine-magnesium exchange using *i*-PrMgCl or *i*-PrMgCl·LiCl. ^c0.3 equiv of Ti(OEt)₄ was used. ^d2.5 equiv. Ar'MgBr and 0.6 equiv. Ti(OEt)₄ were used. ^e3.5 equiv of Ar'MgBr and 0.6 equiv of Ti(OEt)₄ were used.

2, Entries 1-9). Fluoroanilines are a class of deactivated fluorides, and the presence of a free amino group may also lead to C–N formation side reaction;¹² remarkably, 2-fluoroaniline and 3-fluoroaniline underwent smooth couplings based on our present procedure with no C–N forming reaction being observed (Table 2, Entries 10 and 12). The corresponding couplings of the protected fluoroanilines (3-FC₆H₄N=CHPh and 2-FC₆H₄N=CHPh) also proceeded equally well (Table 2, Entries 11 and 13). Similarly, 2-fluorophenol and 4-fluorophenol also participated in the coupling to afford the desired products in 88% and 75% yields respectively (Table 2, Entries 14 and 15). Also, the present C–F bond activation reaction exhibited high functional group compatibility, and the fluorides bearing sensitive functional groups such as ketone, ester, amide, imine, sulfonamide, sulfonate, free carboxylic acid, *etc.* could couple facilely (Table 2, Entries 16-28). Although the similar functional group compatibility has been demonstrated in our previously reported Fe or Co-catalyzed cross couplings of aryl Grignard

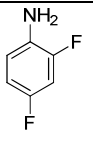
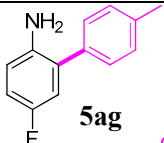
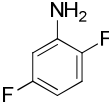
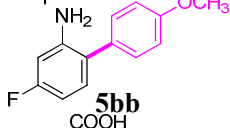
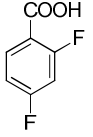
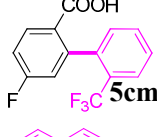
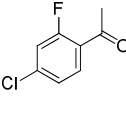
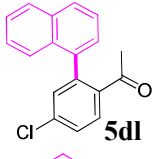
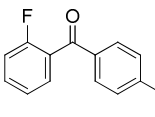
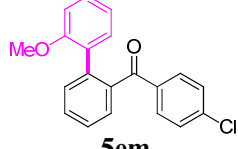
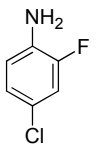
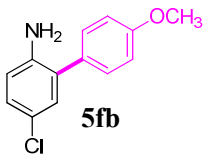
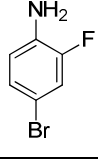
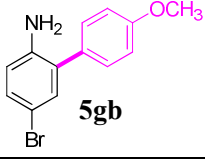
1 reagents in the presence of substoichiometric titanates,^{10,13} those couplings were conducted at -10 °C or
2 room temperature. Such high functional group tolerance was observed for the first time under the heated
3 conditions.
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6 As afore-mentioned, an ketone,^{6d,9} imine (C=NR)^{5b} and ester⁹ functionality usually act as an *ortho*
7 position activating group in C–F bond activation, therefore, the C–F bond cleavage couplings beyond
8 the *ortho*-positions of these groups have been investigated rarely. Our experiments clearly showed that
9 the couplings at *meta* or *para* position to these groups could also take place smoothly (Table 2, Entries
10 16-22). Furthermore, the couplings of two series of 2, 3 or 4-substituted aryl fluorides further
11 demonstrated that the present reactions could proceed well beyond the *ortho* position of an activation
12 group (Table 2, Entries 23-28). Besides, highly functionalized fluorides without an *ortho* group such as
13 ethyl 5-fluoro-2-hydroxybenzoate and ethyl 5-fluoro-2-hydroxy-3-propionylbenzoate could also couple
14 with aryl Grignard reagents to give the cross-coupling products in good yields (Table 2, Entries 29-31).
15 On the other hand, aryl Grignard reagents bearing various sensitive groups such as COOEt (Table 2,
16 Entry 2), CN (Table 2, Entries 3 and 26) and CF₃ (Table 2, Entries 21, 24 and 31) could also undergo
17 this Co-catalyzed coupling. Moreover, heteroaryl Grignard reagents such as 2-thiophenyl (Table 2,
18 Entry 22) and pyridyl (Table 2, Entries 4, 16 and 25) magnesium reagents were also good coupling
19 substrates. Noteworthily, pyridyl fluorides could also undergo the C–F bond activation couplings,
20 leading to the desired pyridine compounds in high yields (Table 2, Entries 32-34).
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23 Selective functionalization of C–F bonds of polyfluorinated molecules is a research area of high interest
24 because it helps to identify the intrinsic factors for cleaving C–F bonds, and it also provides an
25 alternative approach to new fluorocompounds. As illustrated in Table 2 (Entry 8), a selective
26 monoarylation through C–F bond activation of *m*-difluorobenzene was achieved based on our present
27 procedure, leading to 3-phenyl fluorobenzene product in a 65% yield. While *ortho* selective
28 functionalization of C–F bonds of a range of polyfluorinated compounds has been well achieved in the
29 literature utilizing a directing group such as C=NR,^{5b} COR,^{6d,9} 2-oxazoline,^{6e} OH^{4d} and *etc.*, herein we
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have achieved highly *ortho*-selective arylations of difluoroaromatics bearing a free amino or carboxylic group using this Co-catalyzed C–F activation reaction (Table 3, entries 1-3). To the best of our

Table 3 Selective C–F activation couplings in the presence of C–F, C–Cl and C–Br bonds

Entry	ArF	Ar'MgX (Ar')	Product	Yield (%)
1 ^a		4-MeC ₆ H ₄	 5ag	90
2 ^a		4-MeOC ₆ H ₄	 5bb	67
3 ^b		2-F ₃ CC ₆ H ₄	 5cm	81
4 ^{c,d}		1-Naphthyl	 5dl	85
5 ^{c,d}		2-MeOC ₆ H ₄	 5em	88
6 ^{a,d}		4-MeOC ₆ H ₄	 5fb	70
7 ^{a,d}		4-MeOC ₆ H ₄	 5gb	68

^a3.0 equiv of Ar'MgBr and 0.4 equiv. Ti(OEt)₄ were used. ^b2.0 equiv of Ar'MgBr and 0.4 equiv. Ti(OEt)₄ were used.

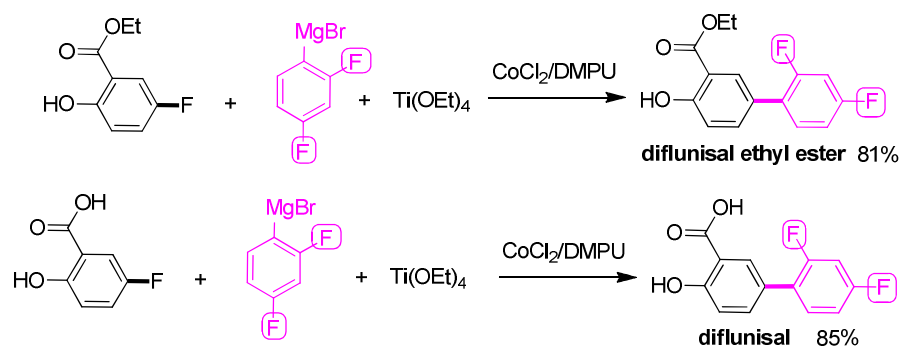
^c1.0 equiv of Ar'MgBr and 0.4 equiv. Ti(OEt)₄ were used. ^dThe diarylated products and those resulting from C–Cl or C–Br vs C–F cleavage were not observed.

knowledge, these were the first examples that a free amino or carboxylic group functioned as an *ortho*-directing group in the Co-catalyzed selective C–F bond activation. Based on these findings, we then investigated the *ortho*-selective C–F bond activation in the presence of C–Cl or C–Br bonds. To

selectively cleave a highly strong bond (C–F) over a relatively weak bond (C–Cl or C–Br) is a remarkably challenging task. Another challenge is that our previous investigation has showed that aryl bromides or aryl chlorides could undergo the cross-couplings with titanate-mediated aryl Grignard reagents under the cobalt catalysis.^{10a} Despite these challenges, the *ortho*-selective C–F bond activation in the presence of C–Cl or C–Br bonds were achieved successfully under the catalysis of CoCl₂/DMPU (Table 3, entries 4-7).

Remarkably, a selective C–F bond activation between two types of C–F bonds on two coupling substrates was also achieved based on this present Co-catalyzed reaction, which was outlined in Scheme 2. In the couplings, the C–F bonds of 5-fluoro-2-hydroxybenzoic acid and its ethyl ester were highly selectively replaced by the aryl group without any activating groups at *ortho* positions, whereas the two fluorine atoms in 2,4-difluorophenylmagnesium bromide were left untouched.¹⁴ It is worth mentioning that the reported methods to yield new fluorinated compound *via* C–F bond cleavage are mainly based on the selective C–F functionalization of polyfluorinated substrates.^{2d} Our selective C–F bond activation illustrated in Scheme 2 provides a new approach to make new fluorine-containing molecules. It can be seen that our couplings facilely afforded an anti-inflammatory agent, **diflunisal**¹⁵ and its ethyl ester in high yields.

Scheme 2 Synthesis of diflunisal and its ethyl ester *via* selective C–F bond activation



Mechanistically, we assume that the cobalt/titanium bimetallic cooperativity plays an important role in the present reaction. As in a tentative transition state (Figure 1), the bimetallic complex promoted the

C–F bond cleavage in a synergetic manner.^{4c,4e,16} Further studies for the cobalt/titanium bimetallic complexes are underway in our laboratories.

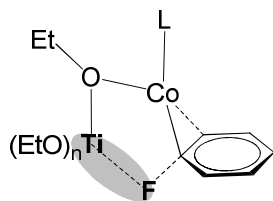


Figure 1 Proposed transition state for the Co/Ti cooperative oxidative addition step

3. CONCLUSION

In conclusion, we have developed an unexpectedly facile Co-catalyzed biaryl cross-coupling reaction *via* C–F bond cleavage. In sharp contrast to the related reported reactions that are dominantly catalyzed by Pd or Ni complexes with NHC or phosphine ligand, and often in the presence of an activating group on the aryl fluorides; the present coupling reaction of C–F bond activation was catalyzed simply by $\text{CoCl}_2/\text{DMPU}$ and proved to be quite general. Remarkably, highly selective C–F activation couplings in the presence of C–F, C–Cl and C–Br bonds could be achieved. We believe that these findings will be helpful to the further development of C–F bond activation reactions and highly effective cobalt catalysts.

4. EXPERIMENTAL SECTIONS

General Information IR spectra were recorded using a FTIR spectrophotometer. ^1H NMR and ^{13}C NMR spectra were recorded on 400 MHz spectrometer (100 MHz for ^{13}C spectroscopy) using TMS as an internal standard. High resolution mass spectra (HRMS) were obtained with a microTOF (ESI). Melting points recorded on a microscopic instrument and are uncorrected.

All reagents and solvents used for aryl magnesium reagents or lithium reagents and reactions were freshly dehydrated and distilled before use. $\text{Ti}(\text{OEt})_4$ was distilled under vacuum before use. The

corresponding glassware was oven dried (120 °C) and cooled under a stream of argon gas. Aryl Grignard reagents such as phenyl magnesium or 4-methoxyphenyl magnesium were prepared according to standard procedure. Pyridyl Grignard reagents were prepared *via* bromine-magnesium exchange using *i*-PrMgCl while functionalized aryl Grignard reagents such as 2-cyanophenylmagnesium chloride or 4-(ethoxycarbonyl)phenyl magnesium chloride were prepared *via* iodine-magnesium exchange using *i*-PrMgCl·LiCl according to Knochel's method.¹⁷ All the Grignard reagents were titrated before use.¹⁸

Typical procedure for Co-catalyzed biaryl couplings *via* C–F cleavage (**3aa**)

Under Ar atmosphere, a solution of PhMgBr in THF (3.0 mmol, 1.0 M in THF) was added dropwise to a solution of Ti(OEt)₄ (273.6 mg, 1.2 mmol) in 10 mL THF under the magnetic stirring at room temperature. After addition, the stirring was continued for 0.5 h.

Under Ar atmosphere, to another three-necked round-bottom flask were added CoCl₂ (19.5 mg, 0.015 mmol), DMPU (38 mg, 0.03 mmol), 2-fluorobenzoic acid ethyl ester (308 mg, 2 mmol) and 10 mL THF. The resulting mixture was heated to 50 °C under the magnetic stirring, and about 2/3 volume of the above-prepared mixture of PhMgBr and Ti(OEt)₄ were added dropwise. After addition, the mixture was stirred at 50 °C for about 3 h. The remaining mixture of PhMgBr and Ti(OEt)₄ were added dropwise and the resulting mixture was stirred at 50 °C until the completion of the reaction. The reaction was quenched with 30 mL distilled water and the product was taken up with CH₂Cl₂ (50 mL × 4). The organic layer was dried over Na₂SO₄ and concentrated to yield the crude compound, which was purified by column chromatography to yield the desired product **3aa** (180 mg, 85% yield).

Note

For the reactions where the fluorinated substrates contain active hydrogen(s) (for example, **3fb**, **3gh**, **3ha**, **3ia**, **3td** and *etc.*), after CoCl₂, DMPU, the fluorinated substrate and THF were added in the flask, one or two equivalents of Ar'MgX was added dropwise to the mixture to neutralize the active hydrogen(s). After that, the mixture was heated slowly to 50 °C. The other operations were conducted as described in typical procedure.

Ethyl biphenyl-2-carboxylate (3aa). The product was prepared as described in typical procedure and isolated as a colorless oil in 85% yield (384 mg); R_f = 0.50 (petroleum ether/ethyl acetate = 20:1, v/v); IR (cm⁻¹, KBr): 1710, 1601, 1258, 698; ¹H NMR (CDCl₃, 400 MHz) δ (ppm): 7.82 (dd, J = 7.6 Hz, J = 0.9 Hz, 1H), 7.55-7.50 (m, 1H), 7.43-7.31 (m, 7H), 4.08 (q, J = 7.1 Hz, 2H), 0.99 (t, J = 7.1 Hz, 3H); ¹³C{¹H}NMR (CDCl₃, 100 MHz) δ (ppm): 168.8, 142.4, 141.6, 134.3, 131.4, 131.1, 130.6, 129.7, 128.4, 128.0, 127.1, 60.9, 13.6. Data was consistent with that reported in the literature.¹⁹

4-Methoxybiphenyl (3bb). The product was prepared as described in typical procedure and isolated as a white solid in a 78% yield (287 mg); mp = 85-86 °C (lit., 85-87 °C); R_f = 0.55 (petroleum ether). IR (cm⁻¹, KBr): 3025, 2918, 1605, 1487, 834, 760, 683; ¹H NMR (CDCl₃, 400 MHz) δ (ppm): 7.59-7.55 (m, 4H), 7.44 (t, J = 6.0 Hz, 2H), 7.34-7.28 (m, 1H), 7.01 (d, J = 6.8 Hz, 2H), 3.88 (s, 3H); ¹³C{¹H}NMR (CDCl₃, 100 MHz) δ (ppm): 159.2, 140.8, 133.8, 128.7, 128.2, 126.8, 126.7, 114.2, 55.4. Data was consistent with that reported in the literature.²⁰

Ethyl biphenyl-4-yl carboxylate (3bc). The product was prepared as described in typical procedure and isolated as a colorless oil in a 55% yield (249 mg); R_f = 0.51 (petroleum ether/ethyl acetate = 20:1, v/v). IR (cm⁻¹, KBr): 3027, 2975, 1705, 1600, 1257, 701; ¹H NMR (CDCl₃, 400 MHz) δ (ppm): 8.12 (d, J = 8.3 Hz, 2H), 7.67-7.62 (m, 4H), 7.49-7.45 (m, 2H), 7.39 (t, J = 7.2 Hz, 1H), 4.40 (q, J = 7.2 Hz, 2H), 1.42 (t, J = 7.1 Hz, 3H); ¹³C{¹H}NMR (CDCl₃, 100 MHz) δ (ppm): 166.5, 145.5, 140.1, 130.1, 129.3, 128.9, 128.1, 127.3, 127.0, 61.0, 14.4. Data was consistent with that reported in the literature.²¹

Biphenyl-2-carbonitrile (3bd). The product was prepared as described in typical procedure and isolated as a colorless oil in a 55% yield (197 mg); R_f = 0.48 (petroleum ether). IR (cm⁻¹, KBr): 3046, 2225, 1599, 1480, 760; ¹H NMR (CDCl₃, 400 MHz) δ (ppm): 7.80 (d, J = 6.2 Hz, 1H), 7.69-7.66 (m, 1H), 7.60 (d, J = 5.8 Hz, 2H), 7.56-7.51 (m, 3H), 7.48 (t, J = 5.6 Hz, 2H); ¹³C{¹H}NMR (CDCl₃, 100 MHz) δ (ppm): 145.5, 138.2, 133.8, 132.9, 130.1, 128.79, 128.77, 127.6, 127.2, 118.8, 111.3. Data was consistent with that reported in the literature.²²

2-Phenylpyridine (3be). The product was prepared as described in typical procedure and isolated as a yellow oil in a 65% yield (202 mg); R_f = 0.58 (petroleum ether/ethyl acetate = 20:1, v/v). IR (cm⁻¹,

KBr): 3056, 1592, 1565, 693; ^1H NMR (CDCl_3 , 400 MHz) δ (ppm): 8.67 (d, $J = 4.8$ Hz, 1H), 7.99-7.97 (m, 2H), 7.67-7.66 (m, 2H), 7.46-7.40 (m, 2H), 7.38-7.36 (m, 1H), 7.17-7.14 (m, 1H); $^{13}\text{C}\{^1\text{H}\}$ NMR (CDCl_3 , 100 MHz) δ (ppm): 157.4, 149.7, 139.4, 136.7, 129.0, 128.8, 126.9, 122.1, 120.5. Data was consistent with that reported in the literature.²³

1-(4-Methoxyphenyl)naphthalene (3db). The product was prepared as described in typical procedure and isolated as a white solid in a 77% yield (360 mg); mp = 116.5-118 °C (lit. 116-117 °C); $R_f = 0.45$ (petroleum ether/ethyl acetate = 10:1, v/v). IR (cm^{-1} , KBr): 3028, 2987, 1601, 1514, 1104, 756; ^1H NMR (CDCl_3 , 400 MHz) δ (ppm): 7.93-7.81 (m, 3H), 7.52-7.39 (m, 6H), 7.04-7.01 (m, 2H), 3.88 (s, 3H); $^{13}\text{C}\{^1\text{H}\}$ NMR (CDCl_3 , 100 MHz) δ (ppm) 159.0, 140.0, 133.9, 133.2, 131.9, 131.1, 128.3, 127.3, 127.0, 126.1, 125.9, 125.7, 125.4, 113.8, 55.4. Data was consistent with that reported in the literature.²⁴

N,N-dimethyl-4-(naphthalen-1-yl)aniline (3df). The product was prepared as described in typical procedure and isolated as a white solid in a 70% yield (346 mg); mp = 108-110 °C. (lit. 109-112 °C); $R_f = 0.40$ (petroleum ether/ethyl acetate = 10:1, v/v). IR (cm^{-1} , KBr): 3045, 2851, 1394, 1352, 1198, 799, 775; ^1H NMR (CDCl_3 , 400 MHz) δ (ppm): 8.01 (d, $J = 8.3$ Hz, 1H), 7.88 (d, $J = 7.7$ Hz, 1H), 7.80 (d, $J = 8.0$ Hz, 1H), 7.51-7.39 (m, 6H), 6.87 (d, $J = 7.9$ Hz, 2H), 3.02 (s, 6H); $^{13}\text{C}\{^1\text{H}\}$ NMR (CDCl_3 , 100 MHz) δ (ppm): 149.8, 140.6, 134.1, 131.0, 129.0, 128.4, 128.1, 127.0, 126.9, 126.5, 125.9, 125.7, 125.6, 112.5, 40.8. Data was consistent with that reported in the literature.²⁵

3-Fluoro-3'-methoxybiphenyl (3eg). The product was prepared as described in typical procedure and isolated as a pale yellow oil in a 65% yield (263 mg); $R_f = 0.58$ (petroleum ether/ethyl acetate = 10:1, v/v). IR (cm^{-1} , KBr): 3073, 2998, 2965, 2938, 2839, 1608, 1587, 700; ^1H NMR (CDCl_3 , 400 MHz): δ (ppm): 7.37-7.32 (m, 3H), 7.28-7.25 (m, 1H), 7.15-7.12 (m, 1H), 7.09-7.08 (m, 1H), 7.04-6.99 (m, 1H), 6.91-6.88 (m, 1H), 3.83 (s, 3H); $^{13}\text{C}\{^1\text{H}\}$ NMR (CDCl_3 , 100 MHz) δ (ppm): 163.4 (d, $J = 243.9$ Hz), 160.3, 143.6 (d, $J = 7.5$ Hz), 141.5 (d, $J = 2.1$ Hz), 130.4 (d, $J = 8.3$ Hz), 130.1, 123.0 (d, $J = 2.4$ Hz), 119.7, 114.3 (d, $J = 21.0$ Hz), 114.1 (d, $J = 21.9$ Hz), 113.4, 113.1, 55.2; ^{19}F NMR (376 MHz, CDCl_3) δ (ppm): -113.1. Data was consistent with that reported in the literature.²⁶

1 *1,3-Bis(4-methoxyphenyl)benzene (4eb)*. The product was prepared as described in typical procedure
2 and isolated as a white solid in a 79% yield (458 mg); mp = 203-205 °C (lit., 203-204 °C); R_f = 0.47
3 (petroleum ether/ethyl acetate = 10:1, v/v). IR (cm⁻¹, KBr): 1685, 1571, 1498, 1321, 759; ¹H NMR
4 (CDCl₃, 400 MHz) δ (ppm): 7.72 (t, J = 1.8 Hz, 1H), 7.58 (d, J = 8.9 Hz, 4H), 7.51-7.46 (m, 3H), 7.00
5 (d, J = 8.8 Hz, 4H), 3.86 (s, 6H); ¹³C{¹H}NMR (CDCl₃, 100 MHz) δ (ppm): 159.2, 141.3, 133.8, 129.1,
6 128.3, 125.3, 125.2, 114.2, 55.4. Data was consistent with that reported in the literature.²⁷

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14 *4'-Methoxybiphenyl-3-amine (3fb)*. The product was prepared as described in typical procedure and
15 isolated as a pale yellow solid in a 79% yield (314 mg); mp = 89-91 °C, (lit., 91 °C); R_f = 0.48
16 (petroleum ether/ethyl acetate = 5/1, v/v). IR (cm⁻¹, KBr): 3367, 3022, 1612, 1523, 1296, 1243, 1032,
17 836, 758, 698; ¹H NMR (CDCl₃, 400 MHz) δ (ppm): 7.49 (d, J = 8.8 Hz, 2H), 7.20 (t, J = 7.8 Hz, 1H),
18 6.96-6.94 (m, 3H), 6.88-6.87 (m, 1H), 6.65-6.63 (m, 1H), 3.84 (s, 3H); ¹³C{¹H}NMR (CDCl₃, 100
19 MHz) δ (ppm): 159.1, 146.7, 142.1, 134.0, 129.6, 128.1, 117.4, 114.1, 113.57, 113.55, 55.3. Data was
20 consistent with that reported in the literature.²⁸

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31 *4'-Methylbiphenyl-2-amine (3gh)*. The product was prepared as described in typical procedure and
32 isolated as a pale yellow oil in a 91% yield (333 mg); R_f = 0.62 (petroleum ether/ethyl acetate = 10/1,
33 v/v). IR (cm⁻¹, KBr): 3381, 1614, 1492, 1295, 820, 748, 733; ¹H NMR (CDCl₃, 400 MHz) δ (ppm): 7.34
34 (d, J = 7.9 Hz, 2H), 7.24 (d, J = 7.8 Hz, 2H), 7.15-7.10 (m, 2H), 6.81 (t, J = 7.4 Hz, 1H), 6.74 (d, J = 8.0
35 Hz, 1H), 3.71 (br, 2H), 2.39 (s, 3H); ¹³C{¹H}NMR (CDCl₃, 100 MHz) δ (ppm): 143.9, 137.1, 136.9,
36 130.8, 129.8, 129.3, 128.7, 128.0, 119.0, 116.0, 21.5. Data was consistent with that reported in the
37 literature.²⁹

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47 *2-Phenylphenol (3ha)*. The reaction was quenched with 2M HCl aqueous solution and the product
48 was isolated as a white solid in an 88% yield (299 mg); mp = 58-59 °C. (lit. 56-58 °C); R_f = 0.49
49 (petroleum ether/ethyl acetate = 10/1, v/v). IR (cm⁻¹, KBr): 3513, 1608, 1511, 825, 740, 688; ¹H NMR
50 (CDCl₃, 400 MHz) δ (ppm): 7.54-7.50 (m, 4H), 7.44 (t, J = 5.4 Hz, 1H), 7.32-7.28 (m, 2H), 7.05-7.02
51 (m, 2H), 5.24 (br, 1H); ¹³C{¹H}NMR (CDCl₃, 100 MHz) δ (ppm): 152.5, 137.2, 130.3, 129.3, 129.23,
52 129.18, 128.2, 127.9, 120.9, 115.9. Data was consistent with that reported in the literature.³⁰

1 *4-Phenylphenol (3ia)*. The reaction was quenched with 2M HCl aqueous solution and the product was
2 isolated as a white solid in a 75% yield (255 mg); mp = 164 °C (lit., 163-165 °C); R_f = 0.55 (petroleum
3 ether/ethyl acetate = 10/1, v/v). IR (cm⁻¹, KBr): 3422, 1613, 1523, 836, 758, 698; ¹H NMR (CDCl₃, 400
4 MHz) δ (ppm): 7.56 (d, J = 6.1 Hz, 2H), 7.51 (d, J = 6.6 Hz, 2H), 7.45-7.42 (m, 2H), 7.34-7.31(m, 1H),
5 6.93 (d, J = 6.7 Hz, 2H), 4.81 (br, 1H); ¹³C{¹H}NMR (CDCl₃, 100 MHz) δ (ppm): 155.1, 140.8, 134.1,
6 128.7, 128.4, 126.7, 115.6. Data was consistent with that reported in the literature.³¹

14 *Ethyl 4-(pyridin-2-yl)benzoate (3je)*. The product was prepared as described in typical procedure
15 and isolated as a white solid in a 72% yield (327 mg); mp = 50.5-51.5 °C (lit. 50.5-52.0 °C); R_f = 0.45
16 (petroleum ether/ethyl acetate = 5/1, v/v). IR (cm⁻¹, KBr): 3056, 2983, 1708, 1607, 1585, 1468, 1363,
17 1281, 1018, 870; ¹H NMR (CDCl₃, 400 MHz) δ (ppm): 8.73 (d, J = 4.6 Hz, 1H), 8.16-8.06 (m, 4H),
18 7.79 (d, J = 3.8 Hz, 2H), 7.31-7.27 (m, 1H), 4.41 (q, J = 7.1 Hz, 2H), 1.42 (t, J = 7.1 Hz, 3H);
19 ¹³C{¹H}NMR (CDCl₃, 100 MHz) δ (ppm): 166.4, 156.2, 149.8, 143.4, 136.9, 130.7, 130.0, 126.8,
20 122.8, 121.0, 61.0, 14.3. Data was consistent with that reported in the literature.³²

31 *Biphenyl-2-yl propanone (3ka)*. The product was prepared as described in typical procedure and
32 isolated as a yellow oil in a 90% yield (378 mg); R_f = 0.35 (petroleum ether). IR (cm⁻¹, KBr): 3060,
33 2975, 2937, 2877, 1681, 1605, 1423, 778, 755, 690; ¹H NMR (CDCl₃, 400 MHz) δ (ppm): 7.45-7.39
34 (m, 2H), 7.38-7.29 (m, 5H), 7.27-7.25 (m, 2H), 2.18 (q, J = 7.3 Hz, 2H), 0.83 (t, J = 7.3 Hz, 3H);
35 ¹³C{¹H}NMR (CDCl₃, 100 MHz) δ (ppm): 207.7, 140.1, 139.7, 138.9, 129.3, 129.1, 127.8, 127.6,
36 126.8, 126.6, 126.4, 35.1, 7.5. Data was consistent with that reported in the literature.³³

45 *Biphenyl-4-yl ethanone (3la)*. The product was prepared as described in typical procedure and
46 isolated as a white solid in a 65% yield (255 mg); mp = 121-122 °C (lit., 121-123 °C); R_f = 0.50
47 (petroleum ether/ethyl acetate = 20:1, v/v). IR (cm⁻¹, KBr): 1682, 1601, 1263, 961, 764; ¹H NMR
48 (CDCl₃, 400 MHz) δ (ppm): 8.04 (d, J = 8.4 Hz, 2H), 7.69 (d, J = 8.4 Hz, 2H), 7.64-7.62 (m, 2H), 7.48
49 (t, J = 7.4 Hz, 2H), 7.40 (t, J = 7.3 Hz, 1H), 2.64 (s, 3H); ¹³C{¹H}NMR (CDCl₃, 100 MHz) δ (ppm):
50 197.7, 145.8, 139.9, 135.9, 129.0, 128.9, 128.2, 127.3, 127.2, 26.7. Data was consistent with that
51 reported in the literature.³⁴

Ethyl 4'-methylbiphenyl-4-yl sulfonate (3mg). The product was prepared as described in typical procedure and isolated as a white solid in a 85% yield (469 mg); mp = 132-133 °C (lit., 131-133 °C); R_f = 0.52 (petroleum ether/ethyl acetate = 10:1, v/v). IR (cm⁻¹, KBr): 1595, 1483, 1358, 808, 794, 636; ¹H NMR (CDCl₃, 400 MHz) δ (ppm): 7.96 (d, J = 8.4 Hz, 2H), 7.74 (d, J = 8.4 Hz, 2H), 7.51 (d, J = 8.1 Hz, 2H), 7.30 (d, J = 7.9 Hz, 2H), 4.17 (q, J = 7.1 Hz, 2H), 2.42 (s, 3H), 1.34 (t, J = 7.1 Hz, 3H); ¹³C{¹H}NMR (CDCl₃, 100 MHz) δ (ppm): 146.6, 138.8, 136.1, 134.5, 129.8, 128.3, 127.5, 127.2, 67.0, 21.2, 14.8. Data was consistent with that reported in the literature.^{10a}

4-(4'-methoxybiphenyl-4-ylsulfonyl)morpholine (3nb).³⁵ The product was prepared as described in typical procedure and isolated as a white solid in a 82% yield (572 mg); mp = 198-199 °C; R_f = 0.48 (petroleum ether/ethyl acetate = 3:1, v/v). IR (cm⁻¹, KBr): 1597, 1485, 1108, 945, 766; ¹H NMR (CDCl₃, 400 MHz) δ (ppm): 7.78 (d, J = 8.4 Hz, 2H), 7.71 (d, J = 8.4 Hz, 2H), 7.56 (d, J = 8.7 Hz, 2H), 7.02 (d, J = 8.7 Hz, 2H), 3.87 (s, 3H), 3.76 (t, J = 4.7 Hz, 4H), 3.04 (t, J = 4.6 Hz, 4H); ¹³C{¹H}NMR (CDCl₃, 100 MHz) δ (ppm): 160.2, 145.6, 132.9, 131.5, 128.5, 128.4, 127.1, 114.6, 66.1, 55.4, 46.0.

3'-(Trifluoromethyl)biphenyl-3-carbaldehyde (3oi). The reaction was post-treated as described in the literature^{10b} and the product was isolated as a yellow solid in a 63% yield (315 mg); mp = 142.5-144 °C. (lit. 143-144 °C); R_f = 0.39 (petroleum ether/ethyl acetate = 10:1, v/v). IR (cm⁻¹, KBr): 3025, 2998, 1692, 1598, 1580, 1336, 1165, 825, 790, 761, 700; ¹H NMR (CDCl₃, 400MHz) δ (ppm): 10.11 (s, 1H), 8.15 (t, J = 1.5 Hz, 1H), 7.93-7.90 (m, 2H), 7.69-7.65 (m, 2H), 7.57-7.45 (m, 3H); ¹³C{¹H}NMR (CDCl₃, 100 MHz) δ (ppm): 192.1, 143.9, 140.6, 137.1, 133.0, 131.1 (q, J = 34.8 Hz, 1C), 129.9, 129.8, 129.5, 129.2, 128.0, 124.8, 123.3, 120.9. Data was consistent with that reported in the literature.³⁶

4-(Thiophen-2-yl)benzaldehyde (3pj). The reaction was post-treated as described in the literature^{10b} and the product was isolated as a yellow solid in a 67% yield (252 mg); mp = 68–69 °C (lit. 69.0–69.5 °C); R_f = 0.52 (petroleum ether/ethyl acetate = 10:1, v/v). IR (cm⁻¹, KBr): 3011, 1701, 1604, 1276, 792; ¹H NMR (CDCl₃, 400 MHz) δ (ppm): 10.0 (s, 1H), 7.90 (d, J = 1.8 Hz, 2H), 7.89-7.78 (m, 2H), 7.77 (d, J = 5.0 Hz, 2H), 7.48-7.46 (m, 1H), 7.41-7.39 (m, 1H), 7.15-7.13 (m, 1H); ¹³C{¹H}NMR (CDCl₃, 100

1 MHz) δ (ppm): 191.5, 142.8, 140.1, 135.1, 130.5, 128.5, 126.9, 126.1, 125.1. Data was consistent with
2 that reported in the literature.³⁷

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5 *N, N*-dimethyl 4'-methoxybiphenyl-2-carboxamide (**3qb**). The product was prepared as described in
6 typical procedure and isolated as a yellow oil in a 75% yield (383 mg); R_f = 0.45 (petroleum ether/ethyl
7 acetate = 1:1, v/v). IR (cm⁻¹, KBr): 3051, 2970, 1624, 1427, 1286, 768, 759; ¹H NMR (CDCl₃, 400
8 MHz) δ (ppm): 7.45-7.34 (m, 6H), 6.93 (d, J = 8.5 Hz, 2H), 3.84 (s, 3H), 2.88 (s, 3H), 2.41 (s, 3H);
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10 ¹³C{¹H}NMR (CDCl₃, 100 MHz) δ (ppm): 171.6, 159.2, 132.39, 132.35, 131.2, 131.1, 130.9, 129.6,
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12 127.4, 127.2, 113.8, 55.2, 38.29, 38.27; HRMS calcd for C₁₆H₁₈NO₂⁺ [M+H]⁺ 256.1332, Found
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14 256.1327.

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21 *N, N*-Dimethyl 3'-(trifluoromethyl)biphenyl-3-carboxamide (**3ri**). The product was prepared as
22 described in typical procedure and isolated as a pale yellow solid in a 76% yield (445 mg); mp = 108-
23 109 °C; R_f = 0.48 (petroleum ether/ethyl acetate = 3:1, v/v). IR (cm⁻¹, KBr): 1597, 1485, 1108, 945, 766;
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29 ¹H NMR (CDCl₃, 400 MHz) δ (ppm): 7.84 (s, 1 H), 7.77 (d, J = 7.6 Hz, 1H), 7.66-7.61 (m, 3H), 7.56 (t,
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31 J = 7.7 Hz, 1H), 7.50 (t, J = 7.6 Hz, 1H), 7.43 (d, J = 7.6 Hz, 1H), 3.14 (s, 3H), 3.03 (s, 3H);
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33 ¹³C{¹H}NMR (CDCl₃, 100 MHz) δ (ppm): 171.1, 141.1, 139.9, 137.2, 131.1 (q, J = 32.0 Hz, 1C),
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35 130.4, 129.4, 129.0, 128.1, 126.4, 125.8, 124.2 (q, J = 3.7 Hz, 1C), 123.8 (q, J = 3.8 Hz, 1C), 122.7,
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37 39.4, 35.2; HRMS calcd for C₁₆H₁₅F₃NO⁺ [M+H]⁺ 294.1100, Found 294.1101.

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41 *N, N*-dimethyl 4-pyridin-3-ylbenzamide (**3sk**). The product was prepared as described in typical
42 procedure and isolated as a white solid in a 76% yield (344mg); mp = 95 °C (lit., 94-96 °C); R_f = 0.32
43 (petroleum ether/ethyl acetate = 1/1, v/v). IR (cm⁻¹, KBr): 1638, 1562, 1395, 1084; ¹H NMR (CDCl₃,
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45 400 MHz) δ (ppm): 8.87 (s, 1H), 8.64 (s, 1H), 7.90 (d, J = 6.2 Hz, 1H), 7.63 (d, J = 6.6 Hz, 2H), 7.55 (d,
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47 J = 6.8 Hz, 2H), 7.41 (s, 1H), 3.15 (s, 3H), 3.05 (s, 3H); ¹³C{¹H}NMR (CDCl₃, 100 MHz) δ (ppm):
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49 171.1, 148.7, 148.1, 138.9, 136.1, 136.0, 134.6, 128.0, 127.1, 123.7, 39.6, 35.4. Data was consistent
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51 with that reported in the literature.^{10a}

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57 2'-Cyanobiphenyl-2-carboxylic acid (**3td**). The reaction was quenched with 2M HCl aqueous
58 solution and the product was isolated as a white solid in a 92% yield (410 mg); mp = 170.4-172.3 °C
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(lit., 170–172 °C); R_f = 0.48 (petroleum ether/ethyl acetate/acetic acid = 5:1:0.01, v/v/v). IR (cm^{-1} , KBr): 3433, 3267, 3051, 2229, 1701, 1602, 1433, 1384, 1095, 692, 521; ^1H NMR (CDCl_3 , 400 MHz) δ (ppm): 8.12 (d, J = 7.4 Hz, 1H), 7.48 (t, J = 7.8 Hz, 1H), 7.35–7.31 (m, 2H), 7.28–7.24 (m, 4H); $^{13}\text{C}\{^1\text{H}\}$ NMR (CDCl_3 , 100 MHz) δ (ppm) 171.0, 135.0, 134.9, 133.7, 132.2, 132.1, 130.2, 130.0, 128.6, 128.54, 128.49, 127.9, 119.2, 117.6. Data was consistent with that reported in the literature.³⁸

Biphenyl-3-carboxylic acid (3ua). The reaction was quenched with 2M HCl aqueous solution and the product was isolated as a white solid in a 86% yield (341 mg); mp = 163–165 °C (lit., 160–161 °C); R_f = 0.32 (petroleum ether/ethyl acetate/acetic acid = 5:1:0.01, v/v/v). IR (cm^{-1} , KBr): 1708, 1693, 1584, 1277, 745, 701; ^1H NMR (CDCl_3 , 400 MHz) δ (ppm): 8.37 (s, 1H), 8.11 (d, J = 7.8 Hz, 1H), 7.85 (d, J = 7.8 Hz, 1H), 7.65 (d, J = 7.4 Hz, 2H), 7.57 (t, J = 7.7 Hz, 1H), 7.48 (t, J = 7.5 Hz, 2H), 7.40 (t, J = 7.3 Hz, 1H); $^{13}\text{C}\{^1\text{H}\}$ NMR (CDCl_3 , 100 MHz) δ (ppm): 171.8, 141.7, 140.0, 132.5, 129.8, 129.02, 128.98, 128.96, 128.9, 127.9, 127.2. Data was consistent with that reported in the literature.³⁹

4-Naphthalen-1-ylbenzoic acid (3vl). The reaction was quenched with 2M HCl aqueous solution and the product was isolated as a white solid in a 78% yield (387 mg); mp = 234–236 °C (lit., 235.2–236.1 °C); R_f = 0.47 (petroleum ether/ethyl acetate/acetic acid = 5:1:0.01, v/v/v). IR (cm^{-1} , KBr): 1685, 1571, 1498, 1321, 759; ^1H NMR (CDCl_3 , 400 MHz) δ (ppm): 8.29–7.89 (m, 5H), 7.66–7.50 (m, 6H); $^{13}\text{C}\{^1\text{H}\}$ NMR (CDCl_3 , 100 MHz) δ (ppm): 171.8, 146.6, 139.0, 133.8, 131.9, 131.7, 131.2, 130.29, 130.27, 128.4, 127.0, 126.4, 126.0, 125.6, 125.3. Data was consistent with that reported in the literature.⁴⁰

Ethyl 4-hydroxy-4'-methoxybiphenyl-3-carboxylate (3wb). The reaction was quenched with 2M HCl aqueous solution and the product was isolated as a white solid in an 86% yield (468 mg); mp = 120–122 °C; R_f = 0.62 (petroleum ether/ethyl acetate = 10/1, v/v). IR (cm^{-1} , KBr): 3138, 1712, 1485, 1439, 1210, 766, 590; ^1H NMR (CDCl_3 , 400 MHz) δ (ppm): 10.81 (s, 1H), 8.01 (d, J = 2.4 Hz, 1H), 7.65 (dd, J = 8.6 Hz, J = 2.4 Hz, 1H), 7.47 (d, J = 8.8 Hz, 2H), 7.03 (d, J = 8.6 Hz, 1H), 6.97 (d, J = 8.8 Hz, 2H), 4.44 (q, J = 7.2 Hz, 2H), 3.85 (s, 3H), 1.43 (t, J = 7.2 Hz, 3H); $^{13}\text{C}\{^1\text{H}\}$ NMR (CDCl_3 , 100 MHz) δ (ppm):

170.2, 160.7, 159.0, 134.0, 132.6, 132.1, 127.7, 127.6, 117.9, 114.3, 112.7, 61.5, 55.3, 14.2; HRMS calcd for $C_{16}H_{17}O_4^+$ $[M+H]^+$ 273.1121, Found 273.1120.

Ethyl 4-hydroxy-4'-methoxy-5-propionyl biphenyl-3-carboxylate (3xb). The reaction was quenched with 2M HCl aqueous solution and the product was isolated as a white solid in a 78% yield (512 mg); mp = 110-112 °C; R_f = 0.45 (petroleum ether/ethyl acetate = 5:1, v/v). IR (cm^{-1} , KBr): 3369, 1685, 1658, 1439, 1251, 766, 699; 1H NMR ($CDCl_3$, 400 MHz) δ (ppm): 12.22 (s, 1H), 8.21 (d, J = 2.5 Hz, 1H), 8.15 (d, J = 2.5 Hz, 1H), 7.51-7.48 (m, 2H), 7.00-6.96 (m, 2H), 4.46 (q, J = 7.2 Hz, 2H), 3.86 (s, 3H), 3.15 (q, J = 7.2 Hz, 2H); 1.44 (t, J = 7.2 Hz, 3H), 1.23 (t, J = 7.2 Hz, 3H); $^{13}C\{^1H\}$ NMR ($CDCl_3$, 100 MHz) δ (ppm): 203.4, 169.1, 160.4, 159.3, 134.1, 133.1, 131.68, 131.64, 127.8, 125.8, 115.7, 114.4, 61.9, 55.4, 35.8, 14.2, 8.3; HRMS calcd for $C_{19}H_{21}O_5^+$ $[M+H]^+$ 329.1384, Found 329.1383.

Ethyl 4-hydroxy-5-propionyl-3'-(trifluoromethyl)biphenyl-3-carboxylate (3xi). The reaction was quenched with 2M HCl aqueous solution and the product was isolated as a white solid in a 73% yield (534 mg); mp = 102-104 °C; R_f = 0.45 (petroleum ether/ethyl acetate = 5:1, v/v). IR (cm^{-1} , KBr): 3398, 1683, 1667, 1447, 1249, 763, 700; 1H NMR ($CDCl_3$, 400 MHz) δ (ppm): 12.3 (s, 1H), 8.26-8.25 (m, 1H), 8.21-8.20 (m, 1H), 7.79-7.74 (m, 2H), 7.63-7.56 (m, 2H), 4.49 (q, J = 7.1 Hz, 2H), 3.17 (q, J = 7.2 Hz, 2H), 1.46 (t, J = 7.1 Hz, 3H), 1.24 (t, J = 7.2 Hz, 3H); $^{13}C\{^1H\}$ NMR ($CDCl_3$, 100 MHz) δ (ppm): 202.7, 169.0, 161.3, 139.9, 134.6, 133.2, 131.3 (q, J = 32.1 Hz, 1C), 130.4, 130.0, 129.4, 126.3, 124.1 (q, J = 3.6 Hz, 1C), 123.4 (q, J = 3.7 Hz, 1C), 122.7, 115.7, 62.1, 36.1, 14.2, 8.2; HRMS calcd for $C_{19}H_{18}F_3O_4^+$ $[M+H]^+$ 367.1152, Found 367.1150.

3-Phenylpyridine (3ya). The product was prepared as described in typical procedure and isolated as a colorless oil in a 81% yield (260 mg); R_f = 0.54 (petroleum ether/ethyl acetate = 5:1, v/v). IR (cm^{-1} , KBr): 3038, 1603, 1521, 699; 1H NMR ($CDCl_3$, 400MHz) δ (ppm): 8.87 (d, J = 1.6 Hz, 1H), 8.61-8.60 (m, 1H), 7.90-7.87 (m, 1H), 7.60-7.58 (m, 2H), 7.49 (t, J = 7.2 Hz, 2H), 7.44-7.36 (m, 2H); $^{13}C\{^1H\}$ NMR ($CDCl_3$, 100 MHz) δ (ppm): 148.4, 148.3, 137.8, 136.7, 134.4, 129.1, 128.1, 127.2, 123.6. Data was consistent with that reported in the literature.⁴¹

Methyl 6-phenylnicotinate (3za). The product was prepared as described in typical procedure and isolated as a white solid in a 92% yield (400 mg); mp = 117-118 °C (lit., 118 °C); R_f = 0.53 (petroleum ether/ethyl acetate = 20:1, v/v). IR (cm⁻¹, KBr): 1716, 1596, 1288, 1117, 752, 698; ¹H NMR (CDCl₃, 400MHz) δ (ppm): 8.71 (s, 2H), 7.64 (d, J = 5.0 Hz, 1H), 7.47-7.39 (m, 3H), 7.35-7.32 (m, 2H), 3.70 (s, 3H); ¹³C{¹H}NMR (CDCl₃, 100 MHz) δ (ppm): 167.3, 151.4, 148.9, 137.7, 137.1, 136.0, 128.5, 128.4, 128.1, 122.5, 52.4. Data was consistent with that reported in the literature.⁴²

5-Fluoro-4'-methylbiphenyl-2-amine (5ag). The product was prepared as described in typical procedure and isolated as a colorless oil in a 90% yield (362 mg); R_f = 0.49 (petroleum ether/ethyl acetate = 10:1, v/v). IR (cm⁻¹, KBr): 3422, 3375, 1712, 1599, 1487, 1241, 1168, 1034, 829, 814, 755, 697; ¹H NMR (CDCl₃, 400 MHz) δ (ppm): 7.33 (d, J = 8.0 Hz, 2H), 7.26 (d, J = 7.6 Hz, 2H), 6.88-6.83 (m, 2H), 6.70-6.67 (m, 1H), 3.59 (s, 2H), 2.40 (s, 3H); ¹³C{¹H}NMR (CDCl₃, 100 MHz) δ (ppm): 156.4 (d, J = 234.7 Hz), 139.6 (d, J = 1.2 Hz), 137.4, 135.6 (d, J = 1.2 Hz), 129.6, 128.8, 116.7 (d, J = 22.2 Hz), 116.4 (d, J = 7.9 Hz), 114.6 (d, J = 22.2 Hz), 21.2; ¹⁹F NMR (376 MHz, CDCl₃) δ (ppm): -127.0. Data was consistent with that reported in the literature.⁴³

4-Fluoro-4'-methoxybiphenyl-2-amine (5bb). The product was prepared as described in typical procedure and isolated as a yellow oil in a 67% yield (291 mg); R_f = 0.31 (petroleum ether/ethyl acetate = 5:1, v/v). IR (cm⁻¹, KBr): 3425, 3380, 1598 1480, 1167, 698; ¹H NMR(400 MHz, CDCl₃) δ (ppm): 7.32 (d, J = 8.7 Hz, 2H), 7.25-7.00 (m, 1H), 6.97 (d, J = 8.7 Hz, 2H), 6.51-6.42 (m, 2H), 3.84 (s, 3H); ¹³C{¹H}NMR (100 MHz, CDCl₃) δ (ppm): 163.0 (d, J = 241.6 Hz), 158.9, 145.2 (d, J = 10.8 Hz), 131.6 (d, J = 9.7 Hz), 130.9, 130.3, 123.3 (d, J = 2.7 Hz), 114.3, 105.0 (d, J = 21.2 Hz), 101.9 (d, J = 24.6 Hz), 55.3; ¹⁹F NMR (376 MHz, CDCl₃) δ (ppm): -135.1. Data was consistent with that reported in the literature.⁴⁴

5-Fluoro-2'-trifluoromethylbiphenyl-2-carboxylic acid (5cm). The reaction was quenched with 2M HCl aqueous solution and the product was isolated as a yellow solid in a 81% yield (460 mg); mp = 164.5-165.4 °C; 81%; R_f = 0.49 (petroleum ether/ethyl acetate/acetic acid = 5:1:0.01). IR (cm⁻¹, KBr): 3421, 3066, 1692, 1600, 1435, 766; ¹H NMR (CDCl₃, 400 MHz) δ (ppm): 8.33 (s, 1H), 7.57-7.53 (m,

3H), 7.46-7.42 (m, 3H); $^{13}\text{C}\{^1\text{H}\}$ NMR (CDCl_3 , 100 MHz) δ (ppm): 167.7, 167.1, 136.5, 134.4, 131.1, 129.7, 129.5, 127.2, 126.4, 125.4, 124.7, 124.1, 115.2, 110.7; ^{19}F NMR (376 MHz, CDCl_3) δ (ppm): -98.7, -104.6. Data was consistent with that reported in the literature.^{10c}

1-(4-Chloro-2-(naphthalen-1-yl)phenyl)ethanone (5dl). The product was prepared as described in typical procedure and isolated as a white solid in a 85% yield (476 mg); mp = 118-120 °C; R_f = 0.52 (petroleum ether/ethyl acetate = 20:1, v/v). IR (cm^{-1} , KBr): 3061, 2998, 2937, 1682, 1608, 1421, 751, 698; ^1H NMR (CDCl_3 , 400 MHz) δ (ppm): 7.92 (d, J = 8.2 Hz, 2H), 7.72 (d, J = 8.4 Hz, 1H), 7.58 (d, J = 8.4 Hz, 1H), 7.54 -7.45 (m, 4H), 7.423-7.418 (m, 1H), 7.34-7.32 (m, 1H), 1.75 (s, 3H); $^{13}\text{C}\{^1\text{H}\}$ NMR (CDCl_3 , 100 MHz) δ (ppm): 201.6, 141.0, 139.4, 137.4, 137.1, 133.6, 131.7, 131.5, 130.1, 128.9, 128.5, 128.1, 127.3, 126.9, 126.3, 125.33, 125.31, 29.7; HRMS calcd for $\text{C}_{18}\text{H}_{14}\text{ClO}^+$ $[\text{M}+\text{H}]^+$ 281.0728, Found 281.0726.

(2'-Methoxy-biphenyl-2-yl) (4-chlorophenyl) methanone (5em). The product was prepared as described in typical procedure and isolated as a white solid in a 92% yield (592 mg); mp = 97-98 °C; R_f = 0.38 (petroleum ether/ethyl acetate = 10:1, v/v). IR (cm^{-1} , KBr): 1659, 1572, 1497, 1319, 758; ^1H NMR (CDCl_3 , 400 MHz) δ (ppm): 7.68 (d, J = 8.5 Hz, 2H), 7.59-7.54 (m, 1H), 7.48-7.38 (m, 3H), 7.30-7.18 (m, 4H), 6.97-6.93 (m, 1H), 6.67 (d, J = 8.2 Hz, 1H), 3.38 (s, 3H); $^{13}\text{C}\{^1\text{H}\}$ NMR (CDCl_3 , 100 MHz) δ (ppm): 195.7, 155.2, 138.7, 138.5, 137.9, 135.7, 131.5, 131.3, 131.0, 130.8, 129.3, 129.1, 128.5, 128.2, 126.8, 121.0, 110.2, 54.4. HRMS calcd for $\text{C}_{20}\text{H}_{16}\text{ClO}_2^+$ $[\text{M}+\text{H}]^+$ 323.0833, Found 323.0831.

5-Chloro-4'-methoxybiphenyl-2-amine (5fb). The product was prepared as described in typical procedure and isolated as a white solid in a 78% yield (326 mg); mp = 83.5-85.5 °C (lit., 84.5-85.5 °C); R_f = 0.43 (petroleum ether/ethyl acetate = 10:1, v/v). IR (cm^{-1} , KBr): 3463, 3372, 1608, 1487, 1241, 1168, 1034, 829, 814, 697; ^1H NMR (CDCl_3 , 400 MHz) δ (ppm): 7.34 (d, J = 8.6 Hz, 2H), 7.09-7.07 (m, 2H), 6.97 (d, J = 8.7 Hz, 2H), 6.71-6.68 (m, 1H), 3.84 (s, 3H); $^{13}\text{C}\{^1\text{H}\}$ NMR (CDCl_3 , 100 MHz) δ

(ppm): 159.1, 142.3, 130.5, 130.1, 130.0, 128.7, 127.9, 123.1, 116.6, 114.4, 55.4. Data was consistent with that reported in the literature.⁴⁵

5-Bromo-4'-methoxybiphenyl-2-amine (5gb). The product was prepared as described in typical procedure and isolated as a white solid in a 68% yield (377 mg); mp = 89-91 °C (lit., 89.1-90.6 °C); R_f = 0.42 (petroleum ether/ethyl acetate = 10:1, v/v). IR (cm⁻¹, KBr): 3458, 3369, 1598, 1477, 1290, 1168, 1034, 814, 697; ¹H NMR (CDCl₃, 400 MHz) δ (ppm): 7.33 (d, J = 8.8 Hz, 2H), 7.21-7.19 (m, 2H), 6.97 (d, J = 8.8 Hz, 2H), 6.63-6.61 (m, 1H), 3.84, (s, 3H); ¹³C{¹H}NMR (CDCl₃, 100 MHz) δ (ppm): 159.1, 142.9, 132.8, 130.7, 130.4, 130.1, 129.2, 117.0, 114.4, 110.1, 55.4. Data was consistent with that reported in the literature.⁴⁶

Ethyl 2',4'-difluoro-4-hydroxybiphenyl-3-carboxylate (diflunisal ethyl ester). The reaction was quenched with 2M HCl aqueous solution and the product was isolated as a pale yellow oil in a 92% yield (450 mg); R_f = 0.55 (petroleum ether/ethyl acetate = 10:1, v/v). IR (cm⁻¹, KBr): 3137, 1679, 1485, 1439, 1210, 766; ¹H NMR (CDCl₃, 400 MHz) δ (ppm): 10.9 (s, 1H), 7.98-7.97 (m, 1H), 7.59 (d, J = 8.7 Hz, 1H), 7.40-7.34 (m, 1H), 7.05 (d, J = 8.6 Hz, 1H), 6.97-6.88 (m, 2H), 4.44 (q, J = 7.2 Hz, 2H), 1.43 (t, J = 7.1 Hz, 3H); ¹³C{¹H}NMR (CDCl₃, 100 MHz) δ (ppm): 170.0, 162.2 (dd, J = 247.2 Hz, J = 11.7 Hz, 1C), 161.3, 159.7 (dd, J = 248.0 Hz, J = 11.7 Hz, 1C), 136.1 (d, J = 2.9 Hz, 1C), 131.1 (dd, J = 9.4 Hz, J = 4.9 Hz, 1C), 130.1 (d, J = 2.7 Hz, 1C), 126.0, 124.2 (dd, J = 13.4 Hz, J = 3.7 Hz, 1C), 117.8, 112.7, 111.6 (dd, J = 21.0 Hz, J = 3.7 Hz, 1C), 104.4 (dd, J = 26.5 Hz, J = 25.1 Hz, 1C), 61.7, 14.2. ¹⁹F NMR (376 MHz, CDCl₃) δ (ppm) -111.5 (d, J = 7.8 Hz, 1F), -113.7 (d, J = 7.4 Hz, 1F). Data was consistent with that reported in the literature.⁴⁷

2',4'-Difluoro-4-hydroxybiphenyl-3-carboxylic acid (diflunisal). The reaction was quenched with 2M HCl aqueous solution and the product was isolated as a white solid in an 87% yield (425 mg); mp = 210-212 °C (lit., 210-211 °C); R_f = 0.33 (petroleum ether/ethyl acetate / acetic acid = 10:1:0.01, v/v/v). IR (cm⁻¹, KBr): 3098, 1682, 1600, 1496, 1467, 1326, 1162, 701; ¹H NMR (CD₃SOCD₃, 400 MHz) δ (ppm): 7.92 (s, 1H), 7.70-7.67 (m, 1H), 7.61-7.55 (m, 1H), 7.39- 7.33 (m, 1H), 7.21-7.16 (m, 1H), 7.08 (d, J = 8.6 Hz, 1H); ¹³C{¹H}NMR (CD₃SOCD₃, 100 MHz) δ (ppm) 171.5, 160.7, 160.4 (dd, J = 245.1

Hz, $J=12.2$ Hz, 1C), 158.9 (dd, $J=246.5$ Hz, $J=12.3$ Hz, 1C), 135.8 (d, $J=2.2$ Hz, 1C), 131.5 (dd, $J=9.5$ Hz, $J=4.6$ Hz, 1C), 130.2 (d, $J=3.1$ Hz, 1C), 125.1, 123.7 (dd, $J=13.2$ Hz, $J=3.7$ Hz, 1C), 117.5, 113.1, 112.0 (dd, $J=20.9$ Hz, $J=3.6$ Hz, 1C), 104.1 (t, $J=26.4$ Hz, 1C); ^{19}F NMR (376 MHz, CDCl_3) δ (ppm): -111.4 (d, $J=7.4$ Hz, 1F), -114.2 (d, $J=7.4$ Hz, 1F). Data was consistent with that reported in the literature.⁴⁸

ASSOCIATED CONTENT

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Notes

The authors declare no competing financial interest.

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Supporting Information

Copies of ^1H NMR, ^{13}C NMR spectra for all products; ^{19}F NMR spectra for the fluorine-containing products and HRMS spectra for the new compounds. This material is available free of charge *via* the Internet at <http://pubs.acs.org>.

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